

Nidhin Gangisetty

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EDUCATION

University of Maryland, College Park

May 2028

B.S. Computer Science, Minors in Robotics and Entrepreneurship, Honors College

GPA: 3.801/4.00

Relevant Coursework: Algorithms, Linear Algebra, Statistics, Computer Systems, Robotics Programming, Discrete Structures, Organization of Programming Languages, Introduction to Robotics, Computer Vision, Data Science

EXPERIENCE

The Aerospace Corporation

May 2026 – Aug. 2026

Software Engineering Intern

Chantilly, VA

- Wrote a **Rust** de-serialization engine to parse **XML**-formatted requests for a mission-critical space system to flag malicious and invalid inputs, outperforming alternative implementations by **35%**
- Deployed **COSMOS** dashboards and **TCP** networking across a custom autonomous drone's signal-classification pipeline, handling **200 packets/sec** across **4 data loggers** with **JSON-to-binary** parsing
- Engineered a **CNN** classifier (BPSK, QPSK, 8PSK) on IQ data with **PyTorch/TorchSig** with **97.8% accuracy**; ran live inference on ground loggers from drone antenna, using **1024-sample** windows visualized with matplotlib

Robotics Algorithms & Autonomous Systems Lab (RAAS)

Oct. 2025 – May 2026

Undergraduate Researcher

College Park, MD

- Adapted the **ITPS framework** (MIT CSAIL) to a **UR3 robotic arm** block-stacking task in MuJoCo, enabling human-guided trajectory steering at **inference time without retraining**
- Implemented and validated all 4 **ITPS steering methods** in 2D simulation, extending support to **CPU and CUDA execution** (Post-Hoc, Diffusion, Sampling, Initialization)

Xprotege Institute of Technology and Management

Sep. 2025 – Dec. 2025

Software Engineering Intern

Ellicott City, MD

- Developed a production-grade **RAG** system using LlamaIndex, Qdrant, and GPT-4 API to automate RFP analysis, increasing throughput and reducing evaluation time by **75%** while maintaining **92%** semantic retrieval accuracy
- Optimized embedding models and retrieval pipelines through adaptive chunk sizing and reranking, enhancing context relevance by **35%** across **200+** page technical documents
- Architected a scalable prompt-engineering framework integrating proprietary scoring logic into LLM outputs, improving decision reliability and boosting bid/no-bid classification accuracy by **40%**

PROJECTS

UR3e Robotic Arm Manipulation — Python, ROS 2, Ubuntu, OpenCV

- Designed a **ROS 2** vision-guided control system for a UR3e robotic arm using pose estimation and inverse kinematics to stack objects with **98% accuracy**, demonstrating real-time manipulation for industrial automation

Kato Apartment Finder — Next.js, Vercel, Supabase, Mapbox, PostgreSQL, Git, Firecrawl — [Website Link](#)

- Launched Kato, an off-campus apartment finder enabling data-driven housing decisions; secured **\$4,000 in awarded funding** and attracted **16,000+ visits** by scraping real-time data from 18 apartments into a unified UI

ACTIVITIES AND LEADERSHIP

Project Lift — Co-Founder, President

May 2025 – Present

- Co-founded a student tech consulting club at UMD, delivered customized tools across **5+ small business clients** and generating **\$50K+** in digital transformation value
- Mentored **50+ interdisciplinary members** in full-stack development and client management, oversaw end-to-end project delivery from requirements gathering to deployment

Alpha Kappa Psi Business Fraternity — Vice President of Professional

Apr. 2025 – Present

- Spearheaded professional development initiatives and corporate outreach for **90+ members**, fostered relationships with **Fortune 500** companies to drive recruitment pipelines and career opportunities

TECHNICAL SKILLS

Languages: Java, Python, C, C++, Rust, JavaScript, TypeScript, SQL, MATLAB

Frameworks, APIs & Databases: React, React Native, Node.js, FastAPI, ROS 2, PostgreSQL, Supabase, AWS (S3, CloudFront, RDS, Lambda), Linux/Unix, TCP/UDP

Libraries & Tools: PyTorch, TensorFlow, NumPy, Pandas, OpenCV, LlamaIndex, Qdrant, Git/GitHub, CAD (Onshape), PyGame, Gazebo, Docker, CI/CD (Github Actions)